Murtaza Majid

Data Science Undergraduate

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Summary

Final-year Data Science student at Air University with 2 internships and 9 deployed ML and AI projects spanning computer vision, NLP, RAG pipelines, and tabular machine learning. Built and shipped production systems serving real users, from a citizen AI assistant (Awam Assist) to a 5-model marketing ML pipeline. Shipped a production MLOps pipeline on AWS EC2 with automated drift detection, GitHub Actions CI/CD, MLflow experiment tracking, and zero-touch Docker deployment.

Education

Bachelor of Science in Data Science **Expected: June 2026**
Air University, Islamabad

Skills

Languages: Python, SQL
ML & DL: Pandas, NumPy, Scikit-Learn, TensorFlow, PyTorch, XGBoost, LSTM, ARIMA, CNNs, NLP, OpenCV, MediaPipe, YOLOv11, Roboflow, SHAP
LLM & RAG: LangChain, ChromaDB, HuggingFace, Groq API, LLaMA 3.3-70B, LangGraph, Prompt Engineering, RLHF, RAG Pipelines
Tools & Platforms: Git, GitHub, AWS, FastAPI, Streamlit, Docker, MLflow, GitHub Actions, Supabase, MySQL, Vercel, Railway, Power BI

Experience

Computer Vision & AI Intern **Dec 2025 – Feb 2026**
Scoutrex, Islamabad *PyTorch, OpenCV, MediaPipe, Roboflow*
- Optimized **PyTorch** models for cricket video analysis under motion blur and varied lighting; implemented **MediaPipe** pose extraction for biomechanical analysis and bowler speed estimation via FPS-normalized displacement.
- Engineered an action phase segmentation pipeline classifying bowling stages (run-up to follow-through) using temporal motion tracking and multi-angle camera synchronization.

LLM Task Trainer **Nov 2025 – Dec 2025**
Turing *Prompt Engineering, RLHF, Tool Use*
- Engineered **system/user prompts** and multi-step task chains to optimize instruction following and integrate external API tools (Search, News, Yelp) using temporal logic.
- Mitigated **hallucinations** and improved model reasoning through rigorous **RLHF** evaluation, ensuring high-accuracy retrieval and context-aware response generation.

Data Science Intern **Jul 2025 – Sep 2025**
ITSOLERA Pvt. Ltd., Islamabad *Python, Pandas, Scikit-Learn, Streamlit*
- Improved time-series forecast accuracy by **15%** using LSTM, ARIMA, XGBoost, and Random Forest models on business metrics.
- Built interactive **Streamlit dashboards** to visualize trends, KPIs, and model predictions for non-technical stakeholders.

Projects

Awam Assist | *LangChain, ChromaDB, HuggingFace, Groq LLaMA 3.3-70B, FastAPI, React*
- Built a **production RAG chatbot** covering 15 Pakistani government services (NADRA, Zakat, BISP, Passport, FBR), deployed on Vercel and Railway with bilingual support and browser-native **Speech-to-Text**.
- Engineered full pipeline: HuggingFace embeddings, ChromaDB vector store, LangChain LCEL chain, and FastAPI backend serving answers grounded in verified government sources.

Campbell's AI Marketing Hub | *XGBoost, Scikit-Learn, FastAPI, Supabase, Groq, React*
- Built a 5-model ML pipeline on 12,545 transactions: KMeans RFM segmentation, two-tier churn prediction (**ROC-AUC 0.84**), and ABSA (**F1 0.75**); used time-based split to eliminate data leakage.
- Deployed **16-endpoint FastAPI** on Railway with Supabase; generated personalized SMS/email/app messages via **Groq LLaMA 3.3-70B** using 7-feature behavioral profiling across **2,041 customers**.

Pakistan Car Price MLOps Pipeline | *XGBoost, FastAPI, Docker, MLflow, SHAP, AWS EC2, GitHub Actions*
- Built end-to-end MLOps pipeline on **7,995 Pakwheels listings**: XGBoost with Optuna tuning (**MAPE 10.85%, R^2 0.82**), SHAP explainability, and temporal drift validation showing **14.87% MAPE degradation** by 2025.
- Automated full retrain-to-deploy cycle via **5-job GitHub Actions pipeline**: drift detection, Optuna retraining, MLflow logging, Docker push to Hub, and zero-touch **SSH deploy to AWS EC2**.

Awards & Certifications

Winner – Air MLX Hackathon 2025: Achieved **RMSE of 0.13** in parking violation risk prediction using multi-table SQL extraction and PCA-based feature reduction.

Certifications: Intermediate Python (DataCamp); Hadoop and Spark (Coursera); British Airways Data Science Simulation (Forage).